

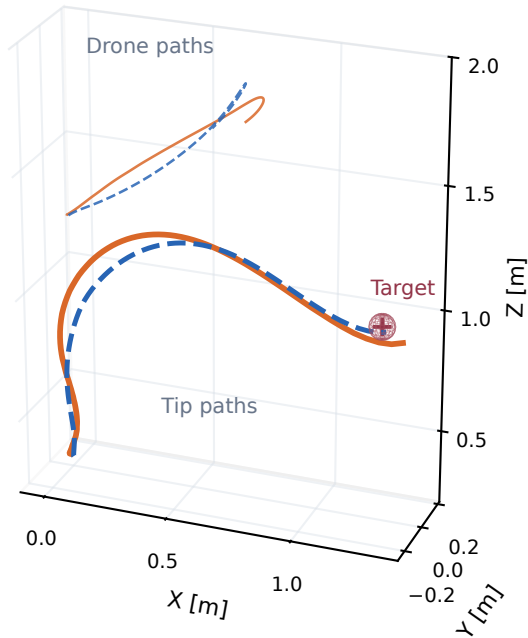
Aerial whip: real flight vs. preflight prediction

M1-full | Real-flight take 004 | MPPI-generated open-loop PVA | 1.20 s whip

— Measured flight - - - Original M1 forecast

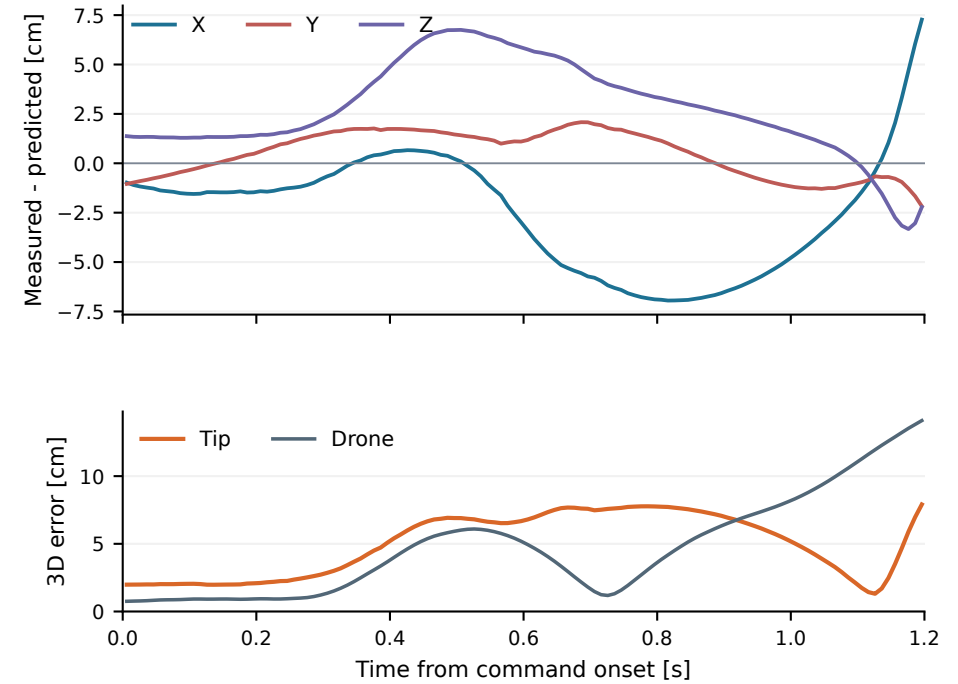
Tip RMS 5.45 cm | Drone RMS 5.94 cm | Tip coverage 100%

A Drone and cable-tip trajectories

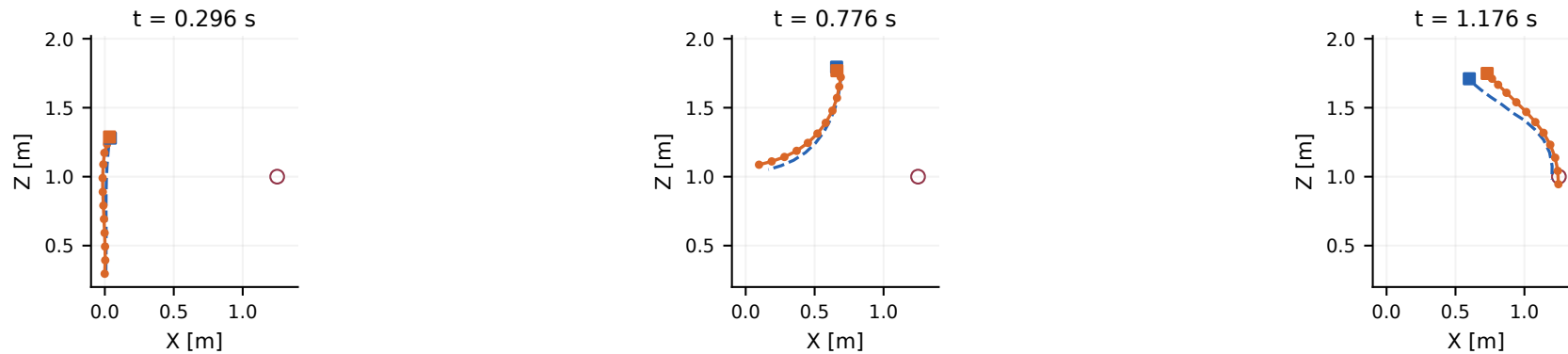


Raw world coordinates. Thin paths: drone; thick paths: tip.

B Tip position residual



C Cable shape at matched times (side view, XZ)



Closest tip approach: 5.87 cm from target centre at 1.177 s; no observed entry into the 5 cm sphere.

Residual = measured - predicted position, not the learned neural correction. Raw XYZ; estimated clock offset; no trajectory recentering.

Selected best-agreement example among 13 development takes, not aggregate validation. Marker gaps are retained; XZ shapes omit lateral error.

Source: whip_m1_004 | Original plan 20260910-181929-716218 | Prediction frozen before flight; no refit or regenerated forecast.